

Analysis

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TPOT with full graph search space versus ensemble

2000 total evals

Version 1

Stats Testing

```
## # A tibble: 24 x 9
##   task_id .y. group1      group2  n1  n2 statistic      p p.signif
##   <int> <chr> <chr>      <chr> <int> <int> <dbl> <dbl> <chr>
## 1   2073 auroc ensemble_random_~ indiv~ 16  16   119 6.29e-3 **
## 2  146818 auroc ensemble_random_~ indiv~ 20  20   160 4 e-2 *
## 3  146820 auroc ensemble_random_~ indiv~ 18  18   149 4.01e-3 **
## 4  168350 auroc ensemble_random_~ indiv~ 20  20   205 1.91e-5 ****
## 5  168757 auroc ensemble_random_~ indiv~ 21  21   149 2.57e-1 ns
## 6  168784 auroc ensemble_random_~ indiv~ 19  19   184 5.34e-5 ****
## 7  168911 auroc ensemble_random_~ indiv~ 3    3    4 7.5 e-1 ns
## 8  190137 auroc ensemble_random_~ indiv~ 19  19   159 8.23e-3 **
## 9  190146 auroc ensemble_random_~ indiv~ 21  21   195 4.28e-3 **
## 10 190411 auroc ensemble_random_~ indiv~ 17  17   139 1.68e-3 **
## 11 359955 auroc ensemble_random_~ indiv~ 21  21   158. 1.49e-1 ns
## 12 359956 auroc ensemble_random_~ indiv~ 19  19   180 1.64e-4 ***
## 13 359958 auroc ensemble_random_~ indiv~ 19  19   183 7.25e-5 ****
## 14 359959 auroc ensemble_random_~ indiv~ 21  21   192 6.28e-3 **
## 15 359960 auroc ensemble_random_~ indiv~ 21  21   21 2.72e-1 ns
## 16 359962 auroc ensemble_random_~ indiv~ 18  18   171 7.63e-6 ****
## 17 359963 auroc ensemble_random_~ indiv~ 19  19   186 2.67e-5 ****
## 18 359964 auroc ensemble_random_~ indiv~ 10  10   44 1.05e-1 ns
## 19 359965 auroc ensemble_random_~ indiv~ 15  15  90.5 1.85e-2 *
## 20 359968 auroc ensemble_random_~ indiv~ 17  17   95 4.04e-1 ns
## 21 359971 auroc ensemble_random_~ indiv~ 3    3    6 2.5 e-1 ns
## 22 359972 auroc ensemble_random_~ indiv~ 5    5   15 6.25e-2 ns
## 23 359974 auroc ensemble_random_~ indiv~ 9    9   44 7.81e-3 **
## 24 359975 auroc ensemble_random_~ indiv~ 15   15   60 1 e+0 ns
```

Medians

```
## # A tibble: 24 x 3
##   task_id individual_tpot ensemble_random_100
##   <int>          <dbl>          <dbl>
## 1   2073          0.879          0.891
## 2  146818          0.936          0.938
## 3  146820          0.996          0.997
## 4  168350          0.966          0.969
## 5  168757          0.791          0.792
## 6  168784          0.966          0.970
## 7  168911          0.871          0.879
## 8  190137          0.925          0.927
## 9  190146          0.963          0.967
## 10 190411          0.923          0.924
## 11 359955          0.738          0.745
## 12 359956          0.935          0.940
## 13 359958          0.951          0.954
## 14 359959          0.746          0.753
## 15 359960          1          1
## 16 359962          0.832          0.842
## 17 359963          0.995          0.996
## 18 359964          0.993          0.993
## 19 359965          1.00          1.00
## 20 359968          0.921          0.921
## 21 359971          0.997          0.998
```

##	22	359972	0.991	0.992
##	23	359974	0.871	0.893
##	24	359975	0.995	0.995